

Topology of the hydrogen bond networks in liquid water at room and supercritical conditions: a small-world structure

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Abstract

Monte Carlo NPT simulations were performed for water at room and supercritical thermodynamic conditions using the TIP5P model potential. The topology of the hydrogen bond patterns was characterized by local (clustering coefficient, path length and degree distribution) and global (spectral analysis) properties. The analysis was performed on 500 uncorrelated configurations at room and supercritical (500 bar and 673 K) conditions. Small-world behavior (highly clustered and small path lengths) was characterized for supercritical water, whereas for room condition, a giant cluster appears and percolates the whole system. These results might have important consequences upon using (near-) supercritical water as new reaction media.

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1. Introduction

Water is the Green Chemistry solvent since it is readily available at low cost, non-toxic, non-inflammable, non-explosive, non-volatile, recyclable and easy to handle. In addition, the physical–chemical properties of liquid water, such as, density, viscosity, dielectric constant, auto-ionization (K_w), ionic and heat conductivity, can be drastically changed or finely tuned by taking it to supercritical or near the supercritical conditions [1]. As a result, water at (near-)supercritical condition has become an important reaction medium where several chemical reactions and synthesis can be performed [2], even without the need for acid–basis catalysts. It has long being established that the physical–chemical properties of water are primarily due to the hydrogen bond (H-bond) network connecting all the water molecules. The behavior of this H-bond network as a function of high temperature and pressure has only recently being probed (revised) by experiments, particularly by

neutron diffraction with isotope substitution [3], which yielded structural information concerning the atom–atom (O–O, O–H and H–H) correlations in supercritical water (673 K and 500 bar vs. $T_c = 647$ K, $p_c = 221$ bar and $\rho_c = 0.32$ g/cm³). However, these are averaged results and a detailed picture of the H-bond network differences for water at room and supercritical conditions is still lacking.

In this Letter, we have used computer simulation (Monte Carlo) to generate liquid structures at room and supercritical conditions from which the topology of the H-bond networks were analyzed in order to establish or not the presence of small-world patterns [4]. This class of disordered networks is widespread in a variety of complex networks [5], such as, the internet topology, neural nets, electronic circuits, food webs, ecosystems, social interactions, scientific collaboration, etc. Due to its apparent universality [5,6], small-world patterns might yield information about the properties of complex networks, such as, evolveability, fitness and robustness with respect to external fluctuations. In the present case it could be, for instance, the stability of the H-bond networks with respect to pressure and temperature

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changes, which might yield new information and perspectives about the relationship between the water properties and the H-bond network.

2. Methodology

The liquid structures were obtained by standard Metropolis Monte Carlo computer simulations [7] using the TIP5P model potential [8] for water-water interaction. The simulations were performed with the DIAD-ORIM program [9] using a constant number of molecules ($N = 300$ and 512), at temperature ($T = 298$ and 673 K) and pressure ($p = 1$ and 500 bar). A cubic simulation box was employed with an initial volume consistent with the experimental density. Periodical boundary conditions, cutoff distance of 900 pm, and a volume variation at every 2000 configurations were used during the simulations. In addition, the range of the molecular translations (maximum translation motion of 15.5 and 57 pm at room and supercritical conditions, respectively), and rotations (maximum rotation motion 15.5° and 57° at room and supercritical conditions, respectively), were adjusted to yield an acceptance ratio of approximately 40% . For equilibration, at least 20×10^6 configurations were generated from simulation boxes already equilibrated in conditions near to the target state. The equilibrium condition has been thoroughly tested using energy, density, radial distribution functions, amongst other properties. Then, the liquid structures used for the H-bond analysis were collected every 3000 equilibrium configurations until it performed 500 simulation boxes. The interval of 3000 configurations (≈ 6 MC cycles, for the large system simulated) has been chosen to guarantee that the liquid structures analyzed were statistically uncorrelated.

The following geometrical restrictions were used in order to establish the presence of an H-bond: $R_{O-O} \leq 330$ pm, for the distance between two oxygen atoms of the water molecules, and $R_{H-X} \leq 274$ pm, for the distance between the hydrogen atom of a donor water molecule and one of the dummy or pseudo-atoms ($X = LP$) that represents the lone-pair (LP) of the acceptor water molecule, in the TIP5P model. These criteria are consistent with those previously proposed [10] using distance and angles. Once the geometrical condition has been satisfied, an additional criterion for the interaction energy of the pair of water molecules is employed, namely, the interaction energy has to be negative (attractive). Finally, these H-bonds are translated into a connectivity matrix that is used in the topological analysis. The topological properties [5] of the H-bond networks were determined by calculating the clustering coefficient C , the path length L , the degree distribution $P(k)$ and the graph spectra $\rho(\lambda)$.

The clustering coefficient, C_i , at node or vertex (molecule) i in the network is defined as the number of existing edges or links (H-bonds) among the k_i neighbors of node i , E_i , divided by the number of possible connections between them, $\binom{k_i}{2}$, that is,

$$C_i = E_i / \binom{k_i}{2} = \frac{2E_i}{k_i(k_i - 1)}. \quad (1)$$

The clustering coefficient of the whole network is the average of all individual C_i 's, namely, $C = \langle C_i \rangle$. It can be formalized in terms of graph theory [11] by considering a set of links $\xi_{ij}(i, j = 1, 2, \dots, N)$, where $\xi_{ij} = 1$ if the nodes i and j are connected, and $\xi_{ij} = 0$ otherwise. Let $\Gamma_i = \{j | \xi_{ij} = 1\}$ be the set of nearest neighbors of node i , with k_i members, then the number of connections between them is

$$E_i = \sum_{j=1}^N \xi_{ij} \left[\sum_{k>j, k \in \Gamma_i}^{k_i} \xi_{jk} \right], \quad (2)$$

and the clustering coefficient at node i can be calculated by Eq. (1). For a random graph, that is, a network with N nodes randomly connected with probability p , the clustering coefficient is [5]

$$C_{\text{rand}} = p = \frac{\langle k \rangle}{N}, \quad (3)$$

where $\langle k \rangle$ is the average degree of the network. It should be emphasized that the nearest neighbors are related to the surroundings of a given node (molecule) in a network (of H-bonds), which does not imply into spatial closeness nor that spatial closeness implies into edges or links (H-bonds). As a result, the average of nearest spatial neighbors of a given node (molecule) should be slightly larger than the average number of connections.

Let $l_{\min}(i, j)$ be the minimum path length (number of edges or links) connecting nodes i and j in the same graph. The average path length of a given node is [11]

$$l(i) = \frac{1}{N} \sum_{j=1}^N l_{\min}(i, j) \quad (4)$$

and the path length is the average over the nodes, namely, $L = \langle l(i) \rangle$. For a random graph the path length scales as [5]

$$L_{\text{rand}} \approx \frac{\ln N}{\ln \langle k \rangle}. \quad (5)$$

A small-world pattern is characterized by $C \gg C_{\text{rand}}$ and $L < L_{\text{rand}}$ [4,5].

Since not all nodes in a network have the same number of edges, a degree distribution $P(k)$ can be defined as the (normalized) frequency of nodes having k edges or node degree. In a random graph the edges are

placed randomly, so the majority of nodes have approximately the same degree, close to the average degree $\langle k \rangle$ of the network. More specifically [5], for a random graph, $P(k)$ is a Poisson distribution peaked at $\langle k \rangle$. Also, for a small-world network described by the Watts–Strogatz model [4], the shape of the degree distribution is similar to that of a random graph [5]. However, for some complex large networks with small-world behavior, the degree distribution significantly deviates from a Poisson distribution, and are usually described as [6]: (i) scale-free networks, with power-law distribution, $P(k) \approx k^{-\gamma}$; (ii) broad-scale networks, $P(k) \approx k^{-\gamma} f(k/k^*)$ where k^* is the cutoff for the sharp cutoff function of the power-law distribution; (iii) single-scale distribution (exponential or Gaussian).

An adjacency matrix $A(G)$ with $N \times N$ elements A_{ij} can represent a graph G with N nodes, when the elements satisfy the following conditions: $A_{ij} = A_{ji} = 1$ if nodes i and j are connected, and 0 otherwise [12]. The diagonalization of the matrix $A(G)$ yields a set of N eigenvalues, λ_j , that is the spectrum of graph G and originates the following spectral density [12]:

$$\rho(\lambda) = \frac{1}{N} \sum_{j=1}^N \delta(\lambda - \lambda_j), \quad (6)$$

which approaches a continuous function as $N \rightarrow \infty$. The k -th moment of the spectral density can be written as the number of paths returning to the same node in the graph, thus yielding some topological features of the network. For a random graph with N nodes and edges generated randomly with probability p , the spectral density converges to a semicircular distribution expressed as [12],

$$\rho(\lambda) = \begin{cases} \frac{\sqrt{4Np(1-p) - \lambda^2}}{2\pi Np(1-p)} & \text{if } |\lambda| < 2\sqrt{Np(1-p)}, \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

The largest (principal) eigenvalue is isolated from the bulk of the spectrum and it decreases as pN .

In this work a FORTRAN program was coded, which reads the coordinates of the water molecules from the simulation boxes, determines the H-bond connections and then perform the topological analysis on the H-bond networks.

3. Results and discussions

Regarding the Monte Carlo simulations the calculated thermodynamic properties agree well with previous simulation results [8], including their dependence with the temperature and pressure. In addition, the calculated radial distribution functions are in good agreement with the experimental data even for supercritical conditions [3]. Table 1 presents the results for the

Table 1
Path length (L and L_{rand}), clustering coefficient (C and C_{rand}) and average number of connections ($\langle k \rangle$) for H-bond networks of water

	298 K and 1 bar		673 K and 500 bar	
	$N = 512$	$N = 300$	$N = 512$	$N = 300$
L	20.36	19.65	3.33	3.30
L_{rand}	5.23	4.90	23.78	31.28
L/L_{rand}	3.9	4.0	0.14	0.11
C	0.5456	0.5302	0.2114	0.2057
$C_{\text{rand}} = p$	0.0064	0.0106	0.0025	0.0041
C/C_{rand}	85.2	50.0	84.5	50.2
$\langle k \rangle$	3.3	3.2	1.3	1.2

clustering coefficient, path length and average number of connections, for 500 H-bond network structures at room and supercritical conditions, obtained from simulations with 300 and 512 water molecules.

For a given thermodynamic condition, it can be observed that the parameters (L and C) determining the local behavior of the H-bond networks, as well as their average degree, are not dependent upon the number of water molecules in the simulation box. This indicates that the simulated samples of 300 and 512 molecules, with periodic boundary conditions in all directions, are large enough to ensure the thermodynamic limit. A comparison between the parameters calculated for different conditions, shows that, at room temperature, the liquid water H-bond networks have $L \gg L_{\text{rand}}$ and $C \gg C_{\text{rand}}$, which suggests that the path length of the network increases with its clustering, characteristic of a giant cluster that percolates the whole system [5]. On the contrary, at the supercritical condition, these networks have much smaller path lengths ($L_{\text{rand}} \gg L$), however the clustering is still large ($C \gg C_{\text{rand}}$), thus yielding the small-world behavior [4] of the H-bond networks at this thermodynamic condition. The average degrees presented in Table 1 agree with the average number of hydrogen bonds, $\langle n_{\text{HB}} \rangle$, of earlier simulations [13]. In particular, it corroborates that at the supercritical condition the average degree ($\langle k \rangle \approx 1.2$ – 1.3) is below the percolation threshold, ~ 1.56 [14], so that a continuous network of H-bonds does not exist, whereas for the room temperature, the percolation persists as can be observed in Fig. 1. For comparison, Fig. 2 illustrates the same H-bond network (upper panel) and its largest cluster (lower panel) at the supercritical condition.

The distribution degree for the H-bond networks of water at room and supercritical conditions are presented in Figs. 3 and 4, respectively. It can be noted that at room temperature it is very similar to a Poisson distribution compatible with a random graph behavior [5]. However, for the supercritical water the distribution is skewed and quite distinct from a Poisson-like behavior. Indeed, it appears similar to broad-scale networks, where the power-law degree has as sharp cutoff [6]. The

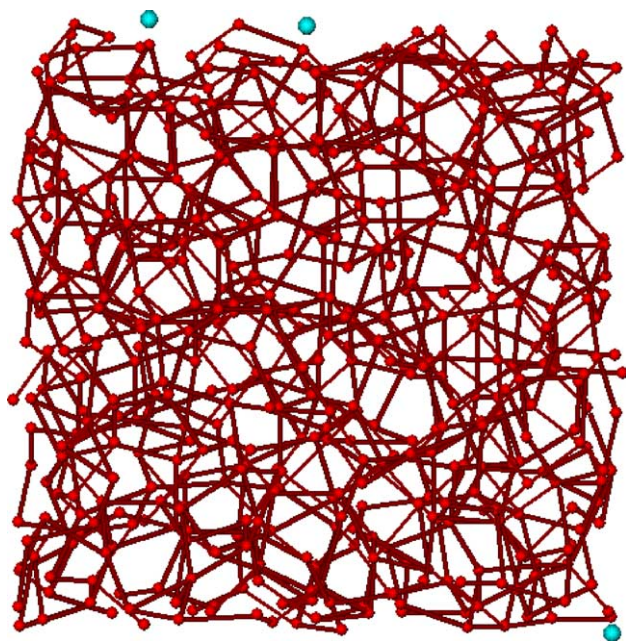


Fig. 1. H-bond network displayed by liquid water at room temperature ($N = 512$). Each node represents a water molecule and the edges the hydrogen bonds.

shape of the distribution has been related, for instance, to constraints limiting the number of links when the connections are costly. This conjecture [6] is consistent to the results observed for H-bond networks of supercritical water, since the rewiring is expensive due to the very low density and long average intermolecular distances, thus relating the shape of the distribution degree to the thermodynamic properties such as density and vaporization enthalpy. It should also be noted that distribution degree is also independent of the number of nodes (300 or 512 molecules).

An analysis using cluster number technique has been performed, but will not be presented and discussed since a similar study is already available [15] for the supercritical water. However, it should be noted that this cluster analysis is quantitatively represented in the averaged local topological properties, such as, clustering coefficients, path lengths and degree distributions.

A global property of complex network is the spectral analysis. The rescaled spectral density of all 500 networks of H-bond for liquid and supercritical water is presented in Fig. 5.

Clearly, the spectral density at room temperature is dominated by the first moment, as expected from the observed percolation process, that is, the formation of giant clusters. On the other hand, for supercritical water, the spectral density has significant contributions from the second and third moments, which is consistent with the small-world behavior of these H-bond networks. Also, it should be noted that when the number of nodes (molecules) increases from 300 to 512 the principal

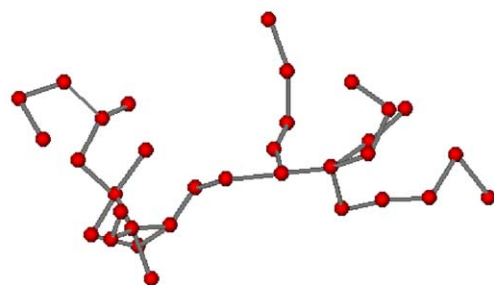
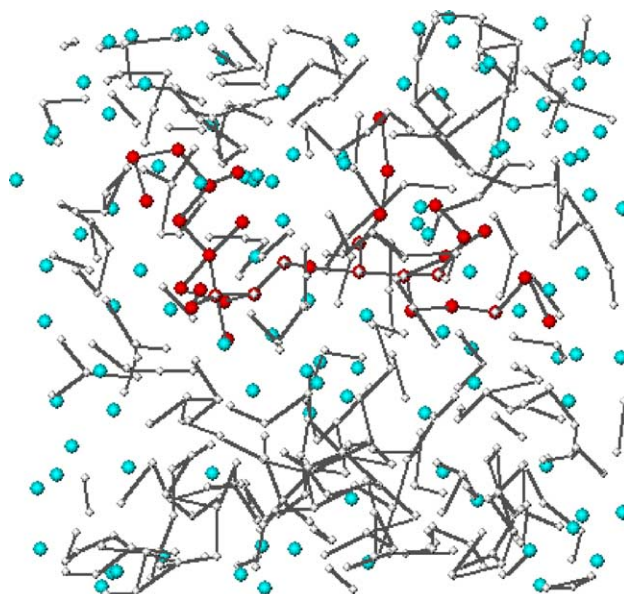


Fig. 2. H-bond network (top panel) and its largest cluster (lower panel) displayed by supercritical water ($N = 512$). Each node represents a water molecule and the edges the hydrogen bonds.

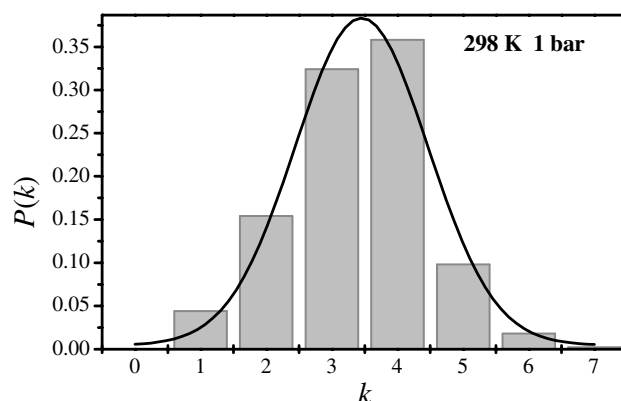


Fig. 3. Degree distribution for the H-bond network of water at room condition.

moment decreases significantly and the spectral density becomes more similar to that for a random graph (semicircular law).

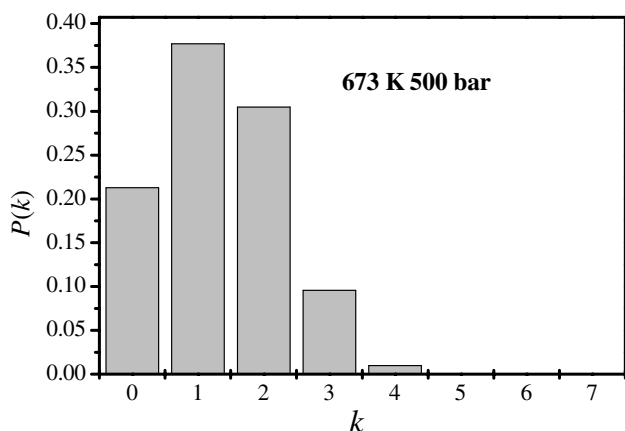


Fig. 4. Degree distribution for the H-bond network of supercritical water.

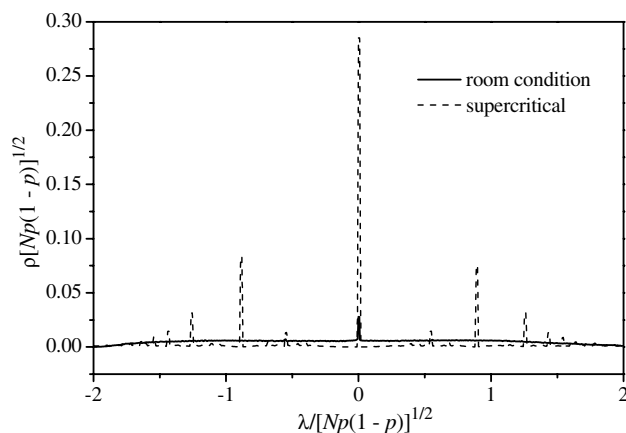


Fig. 5. Rescaled average spectral density for the H-bond networks of water ($N = 512$).

4. Conclusions

The analysis of local topological properties suggests that the H-bond networks in supercritical water present small-world pattern, whereas these networks at room

condition, despite of having large clustering coefficients, do not present small-world behavior, and percolate through the whole system. This distinct behavior has been also clearly observed in the spectral density global property of the connectivity graph. In addition, the local properties (clustering coefficients, path lengths and degree distributions) are independent upon the number of nodes (molecules) for simulation boxes larger than 300 molecules.

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